

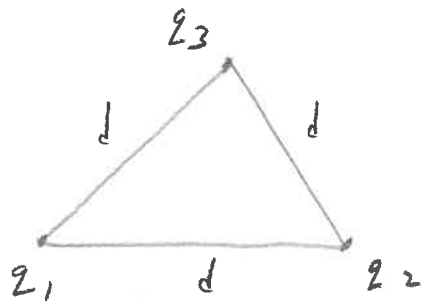
$$C_{hp} + 16 :$$

$$(33) \quad U = \frac{k q_1 q_2}{r} \\ = \boxed{3.75 \times 10^{-14} \text{ J}}$$

$$r = 6.5 \times 10^{-15} \text{ m}$$

$$q_1 = q_2 = e = 1.6 \times 10^{-19} \text{ C}$$

(39)



$$d = 2.5 \text{ cm}$$

$$q_1 = 2 \mu\text{C}, \quad q_2 = 4 \mu\text{C}, \quad q_3 = 6 \mu\text{C}$$

$$U = \frac{k q_1 q_2}{d} + \frac{k q_2 q_3}{d} + \frac{k q_1 q_3}{d} \\ = \frac{k}{d} (q_1 q_2 + q_2 q_3 + q_1 q_3) \\ = \boxed{15.8 \text{ J}}$$

$$(45) \quad \Delta V = 4500 \text{ V}, \quad U = 1.6 \mu\text{J}$$

$$a) \quad U = q \Delta V \rightarrow q = \frac{U}{\Delta V} = 3.56 \times 10^{-10} \text{ C}$$

$$b) \quad n = \frac{q}{e} = \frac{3.56 \times 10^{-10} \text{ C}}{1.6 \times 10^{-19} \text{ C}} = 2.22 \times 10^9 \text{ elementary charges}$$

$$(47) \quad m = 900 \text{ kg}$$

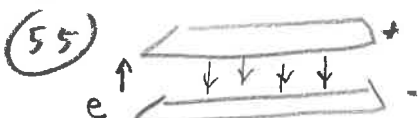
$$\Delta V = 270 \text{ volts}$$

$$Q = 400 \text{ Coulombs}$$

$$U = Q \Delta V = 400 \text{ C} \cdot 270 \text{ Volt} = 108,000 \text{ Joules}$$

$$U = \frac{1}{2} m v_{max}^2$$

$$\rightarrow v_{max} = \sqrt{\frac{2U}{m}} = \boxed{15.5 \text{ m/s}}$$



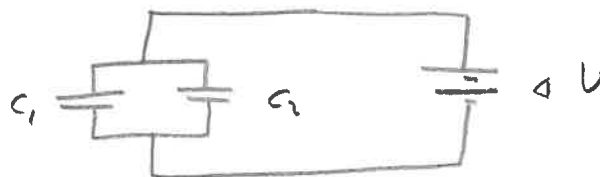
$$E = 850 \frac{\text{V}}{\text{m}}, \quad d = 7.5 \text{ mm}$$

$$\Delta V = E d$$

$$U = q \Delta V = e \Delta V = \boxed{1.02 \times 10^{-18} \text{ J}}$$

$$U = \frac{1}{2} m v_{max}^2 \rightarrow v_{max} = \sqrt{\frac{2U}{m}} \\ = \boxed{1.496 \times 10^6 \text{ m/s}}$$

(65) $C_1 = 0.25 \text{ F}$
 $C_2 = 0.45 \text{ F}$



a) $C_p = C_1 + C_2 = 0.7 \text{ F}$

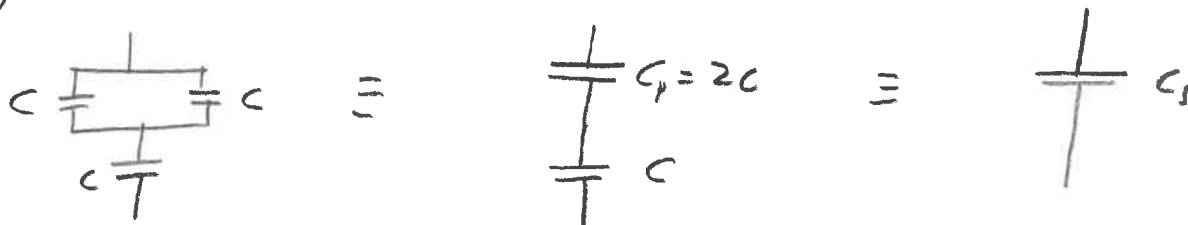
b) $\Delta V = 24 \text{ V}$, $C = \frac{Q}{\Delta V}$

$Q_1 = C_1 \Delta V = 0.25 \text{ F} \times 24 \text{ V} = 6 \text{ Coulombs}$

$Q_2 = C_2 \Delta V = 0.45 \text{ F} \times 24 \text{ V} = 10.8 \text{ Coulombs}$

$Q_{\text{tot}} = Q_1 + Q_2 = 16.8 \text{ Coulombs}$

(73) Three $3 \mu\text{F}$ capacitor



where $\frac{1}{C_s} = \frac{1}{C_p} + \frac{1}{C} = \frac{1}{2C} + \frac{1}{C} = \frac{3}{2C}$

$\rightarrow C_s = \frac{2}{3} C = 2 \mu\text{F}$

(77) $C = 225 \mu\text{F}$
 $\Delta V = 2400 \text{ Volt}$

a) $U = \frac{Q^2}{2C} = \frac{1}{2} C (\Delta V)^2 = \frac{1}{2} (225 \mu\text{F}) 2400 \text{ Volt}$
 $= 648 \text{ J}$

b) Power = $\frac{U}{\Delta t}$, $\Delta t = 2.5 \text{ ms}$
 $= 0.259 \text{ Mega watt}$

(95) $C = 0.45 \times 10^{-9} \text{ F}$
 $A = 1.7 \times 10^{-4} \text{ m}^2$
 $\kappa = 3.8$

$C = \frac{\kappa \epsilon_0 A}{d} \rightarrow d = \frac{\kappa \epsilon_0 A}{C}$
 $= 1.27 \times 10^{-5} \text{ m}$
 $= 0.0127 \text{ mm}$